



**R V COLLEGE OF ENGINEERING**  
(An autonomous institution affiliated to VTU, Belgaum)  
DEPARTMENT OF  
**INDUSTRIAL ENGINEERING & MANAGEMENT**

**IM233AI – WORK SYSTEMS DESIGN**  
**UNIT III**

| Q No | Question                                                                                                    | Marks | Difficulty Level | BT Level | COs |
|------|-------------------------------------------------------------------------------------------------------------|-------|------------------|----------|-----|
| 1.   | Define the Standard Time for a task.                                                                        | 2     | L                | 1        | 3   |
| 2.   | State the fundamental formula used to calculate Standard Time from Normal Time.                             | 2     | L                | 1        | 3   |
| 3.   | Define Normal Performance (or Normal Pace) in the context of time standards.                                | 2     | L                | 1        | 3   |
| 4.   | What is Performance Rating (PR) in time study?                                                              | 2     | L                | 1        | 3   |
| 5.   | List two key prerequisites for determining a valid time standard using the stop-watch method.               | 2     | L                | 1        | 3   |
| 6.   | Why must the work method be standardized before setting a time standard?                                    | 2     | M                | 2        | 3   |
| 7.   | Name the three major types of Allowances (PFD) included in Standard Time.                                   | 2     | L                | 1        | 1   |
| 8.   | Define Personal Needs Allowance and give one example of its use.                                            | 2     | M                | 2        | 1   |
| 9.   | Distinguish between Basic Fatigue Allowance and Variable Fatigue Allowance.                                 | 2     | L                | 2        | 3   |
| 10.  | How is the total Allowance usually expressed when calculating Standard Time?                                | 2     | L                | 1        | 1   |
| 11.  | List the four main methods for determining time standards, ranked by relative accuracy (Highest to Lowest). | 2     | M                | 2        | 3   |
| 12.  | List four main methods for determining time standards, ranked by relative application speed (Fast -Slow).   | 2     | M                | 2        | 3   |
| 13.  | Why is the PMTS method generally considered the most accurate?                                              | 2     | M                | 2        | 3   |
| 14.  | What is the purpose of selecting a "Qualified Worker" at the start of the time study procedure?             | 2     | M                | 2        | 3   |
| 15.  | Define the term Cycle Time in the context of time study.                                                    | 2     | L                | 1        | 1   |
| 16.  | What does the acronym MTM stand for, and what is the unit of time used in MTM-1?                            | 2     | L                | 1        | 3   |
| 17.  | What is the fundamental concept behind the simplified PMTS known as MOST ?                                  | 2     | L                | 1        | 3   |
| 18.  | List the three main sequence models used in Basic MOST.                                                     | 2     | L                | 1        | 3   |
| 19.  | Write the standard formula (sequence) for the General Move sequence in Basic MOST.                          | 2     | L                | 1        | 3   |
| 20.  | What is Mini MOST and for what type of work is it primarily used?                                           | 2     | M                | 2        | 3   |
| 21.  | What is Maxi MOST and when is it typically applied?                                                         | 2     | M                | 2        | 3   |
| 22.  | What is the purpose of MOST for Windows (or electronic PMTS systems)?                                       | 2     | L                | 2        | 3   |



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